

National University of Singapore
School of Computing

Semester 1, AY2026-27

CS4246/CS5446

Reinforcement Learning and Sequential Decision Making

Issued: 02 Sep 2026

Tutorial 3 Week 5: Sequential Decision Making under Uncertainty

Guidelines

You may discuss the content of the questions with your classmates. But everyone should work on and be ready to present ALL the solutions.

Note: Transition Reward Calculation Conventions

AIMA (4e) defines the immediate reward as a *transition reward* $R(s, a, s')$. The reward-on-entry convention is compatible with this definition by taking $R(s, a, s') = f(s')$, that is, reward depends only on the next state. A reward-of-current-state convention corresponds to $R(s, a, s') = f(s)$, where reward depends only on the current state. Both are special cases of AIMA's general $R(s, a, s')$.

In episodic reinforcement learning, terminal states are assigned value 0, and the reward for reaching them is accounted on the transition into the terminal state. This aligns with the Sutton & Barto convention, which is the most common in reinforcement learning.

Both conventions are mathematically valid, but mixing them leads to inconsistent results. It is best to adopt *one convention consistently across all parts of a problem*.

In this course, by default we use the reward-on-entry convention, unless specified otherwise.

Problem 1: Nova the Mars Rover

Nova is an autonomous Mars rover that collects scientific data while managing its limited battery. Each sol, Nova must decide whether to operate in an energy-saving mode or use a high-power boost mode. Boost mode collects more data immediately but drains the battery more quickly. Having learned about Markov Decision Processes (MDPs), Nova's controller models this decision as an infinite horizon MDP with two states—charged and low—and two actions—eco or boost. Therefore, the state set:

$$S = \{\text{charged}, \text{low}\}$$

and action set:

$$A = \{\text{eco}, \text{boost}\}$$

Nova estimates the state-transition probabilities $P(s' \mid s, a)$ as follows:

State s	Action a	$P(s' = \text{charged})$
charged	eco	0.95
charged	boost	0.7
low	eco	0.5
low	boost	0.1

This means that if Nova is charged and uses boost mode, there is a 30% chance its battery becomes low. If Nova is charged and uses eco mode, there is a 95% chance it remains charged. When its battery is low, using eco mode gives a 50% chance of returning to the charged state. However, if Nova uses boost mode while its battery is low, there is only a 10% chance it will become charged again.

Nova estimates the rewards, independent of the resulting state, as follows:

State s	Action a	Reward
charged	eco	7
charged	boost	10
low	eco	0
low	boost	2

Thus, boost mode always gives Nova a higher immediate reward than eco mode. However, Nova performs much better when its battery is charged, and using boost mode makes a low battery more likely.

The challenge is to determine what Nova should do each sol. Nova's decision-making can be modeled as an MDP with two states and two actions, giving $2 \times 2 = 4$ possible policies:

- Always use eco mode.
- Always use boost mode.
- Use eco mode when charged and boost mode when low.
- Use boost mode when charged and eco mode when low.

Without discounting, each of these policies would yield an infinite total reward over an infinite horizon.

Part 1: Discounting

a) Explain the role of discounting future rewards in MDPs. How would an agent behave differently if the discount factor was 0.6 instead of 0.9?

b) As the discount factor varies from 0 to 1, how does the optimal policy change? Provide an example discount factor for each distinct policy obtainable by varying the discount factor.

c) How could the MDP or the discount factor be altered so that the optimal policy is to use eco mode when charged and boost mode when low? Provide a modified MDP that requires minimal changes in probabilities, rewards, or discount factors to achieve this outcome.

d) What does the discount factor γ represent in practical terms? Why can $\gamma = 1$ cause problems?

Part 2: Optimal Policy

Consider Nova's two-state MDP with $\gamma = 0.8$. Represent the value function as $[V_{\text{charged}}, V_{\text{low}}]$. Assume the initial value function is $V^{(0)} = [0, 0]$, and initialize the Q-values as:

$$Q^{(0)}(c, \text{eco}) = 0, \quad Q^{(0)}(c, \text{boost}) = 0, \quad Q^{(0)}(l, \text{eco}) = 0, \quad Q^{(0)}(l, \text{boost}) = 0.$$

a) What are the updated Q-values starting from $V^{(0)}$?

b) What is the next updated value function $V^{(1)}$?

c) Starting from $V^{(1)} = [10, 2]$, compute the next update $V^{(2)}$:

- **Value Iteration (VI):** Apply the Bellman optimality update.
- **Policy Iteration (PI):** Derive the greedy policy $\pi^{(1)}$ wrt $V^{(1)}$, then evaluate it exactly by solving the linear system for $V^{\pi^{(1)}}$.

d) What is the optimal policy?

Problem 2: Aster's Rescue Route—Comparing Active RL Methods

Aster is an autonomous rescue drone learning how to return to base through an uncertain environment. At state s_0 , it can choose between two routes:

$$a = \text{valley route}, \quad b = \text{ridge shortcut}.$$

The drone may reach an intermediate state s_1 or return directly to base T , which is terminal. At s_1 , it can choose between actions c and d .

Assume the environment model is initially unknown. The discount factor is $\gamma = 0.8$.

Before the current episode, Aster has observed the following outcomes from s_0 :

Action	Next state	Reward	Number observed
a	s_1	0	4
b	T	6	3
b	s_1	-2	1

When the previous transition under action b reached s_1 , the episode continued with action d and reward 5 on reaching T .

A new episode now begins. Aster takes action b in s_0 and observes

$$s_0 \xrightarrow{b, -2} s_1.$$

Its exploring behavior policy then selects action d , producing

$$s_1 \xrightarrow{d, 5} T.$$

Consider how different active reinforcement learning methods use this experience.

(a) **Active ADP.**

After observing the new transition $s_0 \xrightarrow{b, -2} s_1$, update the estimated transition probabilities for actions a and b at s_0 .

Suppose the current utility estimates from the learned model are

$$U(s_1) = 4, \quad U(T) = 0.$$

Using the updated model, compute the expected value of actions a and b at s_0 and determine the greedy action.

(b) **Monte Carlo Control.**

Before the new episode, the observed complete returns following (s_0, b) were

$$6, \quad 6, \quad 6, \quad 2.$$

Compute the return following (s_0, b) in the new episode and update the Monte Carlo estimate $Q(s_0, b)$ using all five returns.

(c) **SARSA and Q-learning.**

Consider separate SARSA and Q-learning agents with learning rate $\alpha = 0.5$ and the following current Q-values:

	First action	Second action
s_0	$Q(s_0, a) = 3$	$Q(s_0, b) = 2$
s_1	$Q(s_1, c) = 4$	$Q(s_1, d) = 1$

Both agents observe

$$s_0 \xrightarrow{b, -2} s_1,$$

and the behavior policy selects action d in s_1 .

Compute the updated value of $Q(s_0, b)$ using:

- SARSA;
 - Q-learning.
- (d) Suppose the behavior policy had selected action c rather than d in s_1 . What would happen to the SARSA and Q-learning updates?
- (e) Compare the four active RL methods. What does each learn, and how does each use experience to improve its policy?
- (f) Why is exploration necessary for all four methods?

(Optional) Problem 3: Reinforcement Learning in the Real World

Reinforcement learning is a general computational framework for learning from interaction and feedback. It has important connections to the study of learning and decision making in behavioral psychology and neuroscience, and related ideas arise in evolution and education.

- a. Investigate how reinforcement learning ideas can be used to model human behavior and decision making.
 - b. How is reinforcement learning relevant to one of these fields—evolution and adaptive behavior, behavioral psychology, neuroscience, or education? What connection, if any, exists between innate or biologically grounded reward signals and decision making in that area? How have findings or insights from that area influenced the development of reinforcement learning techniques in machine learning or AI?
-